

Project Report: Food Wastage Management System

1. Introduction and Problem Statement

Food wastage is one of the most pressing environmental and humanitarian challenges of our time. Every day, massive amounts of surplus food from restaurants, grocery stores, and catering services end up in landfills, while simultaneously, countless communities struggle with food insecurity. The primary issue isn't always a lack of food, but rather a lack of communication and logistics between those who have surplus food and the organizations (like NGOs and shelters) that need it.

I built the **Food Wastage Management System (FoodShare)** to solve this exact disconnect. My goal was to create a centralized, data-driven platform that makes it incredibly easy for providers to list their surplus food and for receivers to claim it before it expires.

2. Project Objectives

- **Bridge the Gap:** Create a seamless connection between food donors and receivers.
- **Reduce Spoilage:** Enable real-time tracking of food expiry dates to ensure food is claimed while it's still safe for consumption.
- **Data-Driven Decisions:** Provide actionable insights into where food is being wasted the most, what types of food are in demand, and which cities need the most help.
- **Highlight Environmental Impact:** Translate food rescue metrics into tangible environmental benefits (like CO2 prevented and water saved) to encourage participation.

3. Technology Stack

I chose a modern, Python-centric stack to ensure the application was both robust and visually engaging:

- **Frontend / UI:** Built with **Streamlit**. I went beyond the default Streamlit look by writing and injecting a **Custom CSS Premium Dark Theme**, utilizing modern typography (Inter & Space Grotesk), glassmorphism effects, and micro-animations to give it an enterprise-grade feel.
- **Backend / Database:** Powered by **MySQL**. I designed a relational database schema involving interconnected tables for Providers, Receivers, FoodListings, and Claims.
- **Data Visualization:** Integrated **Plotly Express & Graph Objects** for highly interactive, transparent-background charts that blend seamlessly into the dark UI.

4. Key Features & Architecture

The application is structured into four main operational pillars:

A. Core Operations (CRUD & Navigation)

- **Food Listings & Claims:** Providers can instantly add, update, or delete food listings. Receivers can browse these listings using smart filters (by city, food type, meal type) and place claims.
- **Directory Management:** Complete directories for both Providers and Receivers, allowing users to easily find contact information and addresses.

B. Advanced Analytics & SQL Insights

- **Interactive Dashboards:** Real-time KPI metrics displaying the total food units available, completion rates, and active cities.
- **SQL Data Mining:** I implemented 15 complex SQL queries (using JOINS, GROUP BYs, and aggregations) that run live against the database to answer critical business questions, such as identifying the "Most Claimed Meal Type" and finding "Food Expiring in the Next 30 Days."

C. Pro Evaluation Features (The "Wow" Factor)

To take this project from a standard database app to a real-world product, I implemented three advanced features:

- **Sustainability Impact Dashboard:** A gamified leaderboard that calculates the exact environmental impact of the food saved. For example, it dynamically calculates the kilograms of CO2 emissions prevented and water conserved based on the quantity of completed claims.
- **AI Demand Forecasting & Risk Analysis:** Uses historical claim data to plot a Simple Moving Average (SMA) trendline forecasting future demand. It also scans the database to automatically flag "High-Risk Spoilage Cities" where food listings are high but claim rates are dangerously low.
- **Executive Reporting:** Built-in functionality allowing administrators or auditors to instantly export full snapshots of the Claims and Providers databases directly to CSV files.

5. Challenges and Learnings

One of the biggest challenges was making the UI look sleek and professional. Standard data apps can easily look cluttered. I spent a significant amount of time refining the Custom CSS, ensuring that the metric cards had a subtle glowing border, buttons had smooth gradient hover states, and that all Plotly charts integrated perfectly with the dark theme without clashing white backgrounds.

Additionally, writing complex SQL queries that accurately joined multiple tables to feed the Analytics and AI Forecasting tabs taught me a lot about database optimization and data aggregation.

6. Conclusion

The Food Wastage Management System successfully demonstrates how technology and data analytics can be leveraged to tackle real-world sustainability issues. By combining a robust MySQL backend with a beautifully designed, highly interactive Python frontend, this project not only tracks food logistics but actively encourages waste reduction through gamified environmental metrics and predictive risk analysis.

I am incredibly proud of how this project turned out, both functionally and aesthetically, and I believe it serves as a strong blueprint for how local communities can modernize their food rescue operations.